A Nontrivial Loop at Zero: FinSet K-Theory and Stable Homotopy

N-20251217

§1.1 From the meme to the lecture's explanation

In July 2024, after seeing a meme by Keyao Peng, I read the line

$$0 = 1 - 1 = -1 + 1 = 0$$

through alternating-series regularization and the Eilenberg swindle. A few days before writing this note, while I was reorganizing my homepage, Su Yuyang saw the same line and immediately thought of

$$\sum_{k\geq 0}(-1)^k = 1 - 1 + 1 - 1 + \cdots .$$

That was also my first association. The series has Cesàro and Abel value $1/2$, and it is related to the analytic continuation of the Dirichlet eta function, so the middle expressions naturally look like rearrangement and regularization before they look like topology.

The Hopf-fibration and K-theory clues in the meme led me from analysis to a second mechanism. If an object admits an infinite decomposition

$$S = A \oplus A \oplus A \oplus \cdots ,$$

then

$$S \cong A \oplus S,$$

and a Grothendieck-type invariant may force

$$[A] = 0.$$

This is the Eilenberg swindle: an infinite tail absorbs a finite summand. It gave a coherent explanation of why addition, cancellation, group completion, stability, and vanishing might all sit behind the same arithmetic-looking line.

Keyao Peng's lecture gives the official explanation, and it corrects the object being studied. The two middle expressions are not competing values of an infinite sum, and the equality is not a symbol for absorption at infinity. They are successive frames in a finite motion: create a positive and a negative point, exchange them, and annihilate them again. The comparison is

meme-based infinite-stability reading	lecture's finite-loop explanation
regularization or absorption	creation–swap–annihilation
an infinite object erases a difference	a finite path preserves a distinction
triviality and vanishing	nontriviality of a loop

The more accurate object is therefore not the endpoint 0, but a loop based at 0:

$$\varnothing \sim \{+,-\} \sim \{-,+\} \sim \varnothing.$$

Under the K-theory of finite sets, this loop becomes a nontrivial element. Under the Barratt–Priddy–Quillen theorem, it corresponds to the stable Hopf class.

§1.2 The lecture starts from ordinary equations

The slide deck begins with familiar equations: Pythagoras, Euler's identity, Stokes' theorem, the Navier–Stokes equation, the Langlands slogan, and then the strange meme equation. This is pedagogically important. It suggests that an equation is not merely a string of symbols with a truth value. An equation may also encode a method, a structure, or a path.

In elementary arithmetic,

$$0 = 1 - 1 = -1 + 1 = 0$$

is boring. Both sides are zero. But the lecture asks us to stop looking only at the value. In a higher structure, there can be more than one way for an object to equal itself. The identity loop at 0 and the creation–swap–annihilation loop at 0 need not be the same.

Remark 1.2.1 — This is the conceptual bridge to homotopy type theory and higher category theory. A statement $a = b$ can be represented by a path from a to b. Then two proofs of equality may themselves be compared by a homotopy. The lecture uses this philosophy in a concrete topological model rather than as formal type theory.

§1.3 The Hopf fibration as the first nontrivial map

The first mathematical anchor is the Hopf fibration. Define

$$S^3 = \{(z_1, z_2) \in \mathbb{C}^2 : |z_1|^2 + |z_2|^2 = 1\}.$$

The Hopf map is

$$H : S^3 \to \mathbb{CP}^1 \cong S^2, \qquad (z_1, z_2) \mapsto [z_1 : z_2].$$

For every point of S^2, its preimage is a circle. These circles are linked in a highly organized way.

The Hopf fibration matters because it is not null-homotopic. It represents a nontrivial element of

$$\pi_3(S^2).$$

In beginner language: even though S^3 and S^2 are both spheres, the map $H : S^3 \to S^2$ cannot be continuously shrunk to a constant map.

Definition 1.3.1 (Null-homotopic map)**.** A map $f : X \to Y$ is null-homotopic if it is homotopic to a constant map. Equivalently, there is a continuous family of maps $f_t : X \to Y$ with $f_0 = f$ and f_1 constant.

The Hopf map is the first warning: a process can be topologically nontrivial even if it looks, from far away, like a map between simple spaces.

§1.4 Homotopy groups and the first stable class

For a based space X, the group $\pi_n(X)$ is made from based maps $S^n \to X$, modulo based homotopy. The lecture reviews the low-dimensional intuition:

$$\pi_0(X) = \text{connected components}, \qquad \pi_1(X) = \text{loops modulo homotopy}.$$

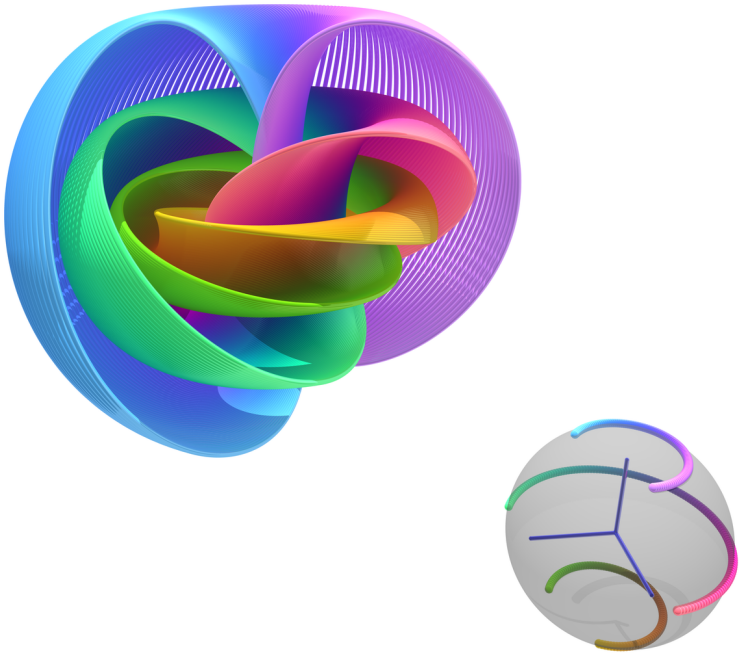


Figure 1.1: the Hopf fibration visualized by linked fibres.

Higher homotopy groups generalize this to maps from higher-dimensional spheres.

For spheres, the groups are easy below the diagonal:

$$\pi_i(S^n) = 0 \qquad (i < n).$$

On the diagonal, one has the degree:

$$\pi_n(S^n) \cong \mathbb{Z}.$$

But above the diagonal the world becomes subtle. The Hopf map gives the famous example

$$\pi_3(S^2) \neq 0.$$

The stable homotopy groups of spheres are obtained by repeatedly suspending:

$$\pi_i^s(\mathbb{S}) = \lim_{n\to\infty} \pi_{i+n}(S^n).$$

The Hopf map contributes the first stable nontrivial class

$$\eta \in \pi_1^s(\mathbb{S}) \cong \mathbb{Z}/2.$$

This class has order two: doing it twice becomes trivial, but doing it once is not trivial.

§1.5 From natural numbers to finite sets

The lecture then changes viewpoint. Instead of defining the natural numbers as primitive symbols, define them as isomorphism classes of finite sets:

$$\mathbb{N} \cong \pi_0(\mathrm{FinSet}).$$

Here a finite set of size n represents the number n. Disjoint union represents addition:

$$A + B := A \sqcup B.$$

To get the integers, one takes a group completion. Informally, one allows formal differences

$$A - B.$$

Thus the number $1 - 1$ is represented by a pair consisting of one positive point and one negative point.

Remark 1.5.1 — The corresponding slide phrases this as a move from numbers to finite sets: the natural number n is represented by an n-element set, and addition is represented by disjoint union. This is the first categorification step in the lecture: instead of remembering only the cardinality, one keeps the finite set itself and later also its symmetries.

Definition 1.5.2 (Grothendieck group)**.** Given a commutative monoid M, its Grothendieck group $K_0(M)$ is the universal abelian group receiving a monoid homomorphism from M. For finite sets under disjoint union this gives

$$K_0(\mathrm{FinSet}) \cong \mathbb{Z}.$$

So far, this still only explains the value of $1 - 1$. It does not yet explain why the path

$$0 \to 1 - 1 \to -1 + 1 \to 0$$

can be interesting. For that, we must remember not just the set of isomorphism classes, but the space of finite sets and isomorphisms.

§1.6 The space of finite sets remembers its symmetries

A set and its cardinality are different amounts of information. Let $\mathcal{F}$ be the *groupoid* of finite sets and bijections. All finite sets of size n are isomorphic, but each such set has an automorphism group Σ_n. Taking the classifying space (the realization of the nerve) gives

$$\boxed{B\mathcal{F} \simeq \coprod_{n \geq 0} B\Sigma_n.}$$

Here $B\Sigma_n$ has one connected component and fundamental group Σ_n. A loop in it remembers an automorphism—an actual permutation of n points—rather than only the number n. In particular,

$$\pi_0(B\mathcal{F}) \cong \mathbb{N}, \qquad \pi_1(B\Sigma_2) \cong \Sigma_2 \cong \mathbb{Z}/2.$$

The crucial distinction is that the empty set belongs to the $n = 0$ component, whose automorphism group is trivial. The transposition loop initially lives in the component of a *two-point* set, not at the empty set. Disjoint union gives $B\mathcal{F}$ the compatible, homotopy-commutative multiplication needed to pass from $\mathbb{N}$ to $\mathbb{Z}$ *without forgetting those loops.* To obtain a loop based in the zero component, we must perform group completion *at the level of spaces*, not merely take the Grothendieck group of the set of cardinalities.

§1.7 The K-space of finite signed sets

Let $K(\mathcal{F})$ denote the E_∞ group completion of $B\mathcal{F} = \coprod_{n\geq 0} B\Sigma_n$ under disjoint union. Ordinary group completion of cardinalities gives only

$$\pi_0 K(\mathcal{F}) \cong \mathbb{Z}.$$

The space-level construction retains homotopies and automorphisms as well. Its correct mathematical input is the groupoid of finite sets and bijections; a model made from signed point configurations visualizes the inverse objects but requires coherent rules for moving and cancelling points. Merely writing $\{+,-\} = \varnothing$ as a formal relation would not determine a homotopy type.

In the signed-configuration picture used by the lecture, the empty configuration is a base point. A positive and a negative point can be created as a cancelling pair, exchanged, and annihilated. This provides a concrete representative of a loop in the zero component of the group-completed space.

The Barratt–Priddy–Quillen theorem, in a precise infinite-loop-space form, identifies this group completion with the underlying infinite-loop space of the sphere spectrum:

$$\boxed{K(\mathcal{F}) \simeq Q(S^0) = \operatorname*{colim}_n \Omega^n S^n.}$$

One useful equivalent description is

$$Q(S^0) \simeq \mathbb{Z} \times (B\Sigma_\infty)^+,$$

where the superscript $+$ denotes the plus construction. In particular,

$$\boxed{\pi_1 K(\mathcal{F}) \cong \pi_1^s(\mathbb{S}) \cong \mathbb{Z}/2.}$$

The group completion has gained negative *components* and kept a nontrivial *loop*. These are two different operations on different layers of information.[1]

§1.8 The loop h

Now the lecture constructs the loop

$$h : \varnothing \sim \{+,-\} \sim \{-,+\} \sim \varnothing.$$

It has three stages.

First, create a cancelling pair:

$$\varnothing \longrightarrow \{+,-\}.$$

Second, swap the two points:

$$\{+,-\} \longrightarrow \{-,+\}.$$

Third, annihilate the cancelling pair:

$$\{-,+\} \longrightarrow \varnothing.$$

[1] For the group-completion and plus-construction descriptions, see Stefan Schwede, *Global Algebraic K-Theory*, Section 8.

K-groups of finite sets

Let $K_i(\mathrm{FinSet}) = \pi_i(K(F), \emptyset)$ to be the higher K-groups, we have the following observations:

- $\pi_0 K(F) = \mathbb{Z}$, by $x \mapsto \|x\|$. This coincides with $K_0(\mathrm{FinSet})$.
- We can construct a loop $h \in \pi_1(K(F), \emptyset)$ by using identifications

$$h : \emptyset \sim \{+, -\} \sim \{-, +\} \sim \emptyset$$

- In fact $\pi_1(K(F), \emptyset) = \mathrm{S}_n^{ab} = \mathbb{Z}/2$ with h as the generator. That is to say, h is a non-trivial identification.

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Figure 1.2: From the lecture slides: the loop h represented by creation, swap, and annihilation.

This is the corrected meaning of

$$0 = 1 - 1 = -1 + 1 = 0.$$

It is not the claim that zero equals zero. It is the claim that there is a particular loop at zero. The middle step, the swap, is the nontrivial part.

Remark 1.8.1 — If the two points were not moved, the movie would look like a boring creation followed by immediate annihilation. The swap forces a braid-like motion. In configuration space, this motion wraps around the collision locus where the two points would meet.

The slides record the key computation:

$$K_1(\mathrm{FinSet}) \cong \mathbb{Z}/2,$$

and the loop h gives the nontrivial element. Doing the swap twice is null-homotopic, but doing it once is not.

§1.9 The parity of a swap is what survives

Why should the exchange of two points represent the nontrivial class? Before group completion there is already an elementary invariant that detects it. Every permutation can be expressed as a product of transpositions. Define

$$\mathrm{sgn} : \Sigma_n \to \{\pm 1\}, \qquad \mathrm{sgn}(\sigma) = (-1)^{\#\text{ transpositions}}.$$

Parity is independent of the chosen expression: for example, it equals the sign of the corresponding permutation matrix's determinant. Thus

$$\operatorname{sgn}(\sigma\tau) = \operatorname{sgn}(\sigma)\operatorname{sgn}(\tau).$$

A single swap has sign -1; two successive swaps have sign $+1$.

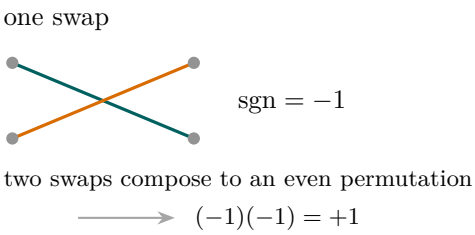


Figure 1.3: A transposition has odd parity, while its square is even. In the stabilized finite-set construction this parity detects the first nontrivial homotopy class.

There is a clean group-theoretic check. For $n \geq 2$, all transpositions in Σ_n are conjugate. Consequently they have the same image in any *abelian* quotient; that common image has order at most two because a transposition squares to the identity. Since transpositions generate Σ_n, its abelianization has at most two elements. The sign map is a surjection onto $\mathbb{Z}/2$, so

$$\boxed{\Sigma_n^{\mathrm{ab}} \cong \mathbb{Z}/2 \qquad (n \geq 2).}$$

Adding an untouched extra point sends Σ_n into Σ_{n+1}, and this stabilization preserves the sign.

The sign calculation alone does not prove the Barratt–Priddy–Quillen theorem or identify *every* higher homotopy group. It does make the first loop intelligible: the nontrivial permutation class survives stabilization and appears, via the theorem, as the nonzero element of

$$\pi_1 K(\mathcal{F}) \cong \mathbb{Z}/2.$$

The lecture's creation–swap–annihilation movie represents this class in a signed-configuration model. The identity movie gives zero; the one-swap movie does not; concatenating two copies gives zero again.

There is a key difference from the Eilenberg swindle. An infinite absorber can make an additive Grothendieck class vanish, but group completion of a homotopy-coherent space remembers the automorphisms of its finite objects. The cancellation of cardinalities has not cancelled the parity of their permutations.

§1.10 Finite sets and stable spheres

The group-completion model above describes an *infinite loop space*. The underlying theorem concerns the corresponding *spectrum*, and this distinction is worth making explicit.

Theorem 1.10.1 (Barratt–Priddy–Quillen)

The connective K-theory spectrum of the symmetric monoidal groupoid of finite sets and bijections (with disjoint union) is equivalent to the sphere spectrum $\mathbb{S}$. In particular,

$$K_i(\mathrm{FinSet}) \cong \pi_i^s(\mathbb{S}) \qquad (i \geq 0).$$

At the bottom of the spectrum we have

$$K_0(\mathrm{FinSet}) \cong \mathbb{Z}, \qquad K_1(\mathrm{FinSet}) \cong \mathbb{Z}/2.$$

The first group keeps track of signed cardinality, while the second remembers the parity detected above. Under the equivalence, the nontrivial one-swap class becomes

$$\eta \in \pi_1^s(\mathbb{S}) \cong \mathbb{Z}/2,$$

the stable Hopf element.

There is a further geometric interpretation, but one should state it at the correct level. By the Pontryagin–Thom theorem, stable homotopy groups of spheres are framed cobordism groups,

$$\pi_i^s(\mathbb{S}) \cong \Omega_i^{\mathrm{fr}}.$$

For $i = 1$, a nonbounding stable framing on a circle represents the nontrivial class. The creation–swap–annihilation picture is a convenient representative of the corresponding finite-set loop; its passage to framed cobordism is justified by the stable homotopy equivalence, not by declaring the naive space of all signed configurations to be literally equal to a framed configuration space.

§1.11 Relation with the classical Hopf map

The classical Hopf fibration represents a nontrivial element of $\pi_3(S^2)$. Stabilizing the Hopf map gives the stable element

$$\eta \in \pi_1^s(\mathbb{S}).$$

The finite-set loop h gives the same class under the BPQ theorem.

Thus the meme ladder is not merely free association. The chain is:

$$\begin{gathered}
\text{Hopf fibration } S^3 \to S^2 \\
\Downarrow \\
\text{stable Hopf element } \eta \in \pi_1^s(\mathbb{S}) \\
\Downarrow \\
K_1(\mathrm{FinSet}) \cong \pi_1^s(\mathbb{S}) \\
\Downarrow \\
\text{the loop } h : \varnothing \sim \{+,-\} \sim \{-,+\} \sim \varnothing.
\end{gathered}$$

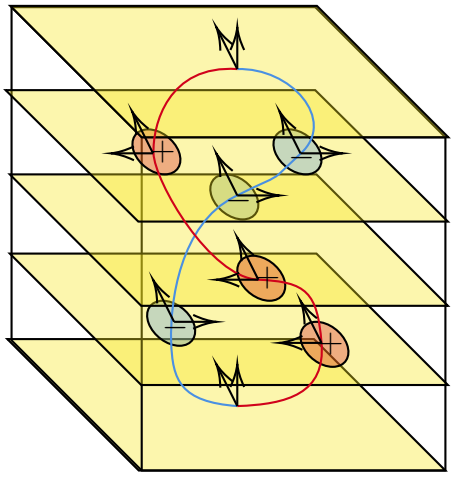
This is the clean mathematical sense in which the arithmetic-looking line is the Hopf map in disguise.

§1.12 Cobordism: the movie becomes a circle

The slides then place the motion in three-dimensional space. As time evolves, the moving point configuration sweeps out a one-dimensional manifold. The creation of the pair, their swap, and their annihilation form a closed loop. Because the points carry signs or framings, the resulting circle is a framed one-manifold.

What about cobordism?

If we place this motion in 3-dimensional space, we obtain a circle:



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Figure 1.4: From the lecture slides: the creation–swap–annihilation movie becomes a framed circle.

Framed cobordism asks whether this framed circle is the boundary of a framed disk. The answer is no for the loop h. This is another expression of the same nontrivial $\mathbb{Z}/2$-class.

Definition 1.12.1 (Framed cobordism, informal)**.** A framed n-manifold is an n-manifold equipped with a trivialization of its normal bundle. Two framed manifolds are framed cobordant if they form the boundary of a framed manifold one dimension higher.

This is the geometric face of the equation. The path starts and ends at the empty configuration, but its trace is not the boundary of a trivial framed disk. The loop is geometrically nonzero.

§1.13 Algebraic K-theory of a field

After finite sets, the lecture repeats the construction with finite-dimensional vector spaces over a field k. Let

$$\mathcal{V}_k = \mathrm{FinVect}_k/\sim$$

be the moduli space of finite-dimensional vector spaces. Under direct sum, it behaves like a categorified version of addition.

The low K-groups are familiar:

$$K_0(k) \cong \mathbb{Z}, \qquad K_1(k) \cong k^\times.$$

The first says that a vector space is measured by its dimension after group completion. The second says that the determinant detects the abelianized automorphism information:

$$\mathrm{GL}_\infty(k)^{\mathrm{ab}} \cong k^\times.$$

This parallels the finite-set case:

object	K_0	K_1
FinSet	$\mathbb{Z}$	$\mathbb{Z}/2$
$\mathrm{FinVect}_k$	$\mathbb{Z}$	$k^\times$

The lecture now asks for a geometric model of these algebraic K-groups beyond degree one.

§1.14 Intrinsic skein-lasagna

The advanced second half of the slide deck introduces intrinsic skein-lasagna. This is not merely decorative. It is a proposed geometric language for K-groups analogous to how framed cobordism gives a geometric language for stable homotopy.

The first definition is an n-treefold: a space locally homeomorphic to

$$T \times \mathbb{R}^{n-1},$$

where T is a tree. The branching models a controlled kind of singularity.

Given a symmetric monoidal category $(\mathcal{C}, \otimes)$, an n-$(\mathcal{C}, \otimes)$-lasagna is, roughly, a framed n-treefold with labels from $\mathcal{C}$, satisfying compatibility at the branching loci.

Definition 1.14.1 (Intrinsic skein-lasagna, informal)**.** An intrinsic lasagna is a geometric object built from treefold-like pieces, with labels coming from a symmetric monoidal category. The word "intrinsic" means that the object is studied on its own, rather than as something embedded in a previously chosen ambient manifold.

The slide illustrates n-treefolds as spaces locally modelled on $T \times \mathbb{R}^{n-1}$, where T is a tree. For the notebook, the useful point is that branching is allowed but controlled.

The word "intrinsic" is important. Classical skein-lasagna modules often involve embedded objects inside an ambient manifold. Here the lecture emphasizes that one can study the geometry of the lasagna itself, without first choosing an ambient space. If desired, one may imagine the ambient space as $\mathbb{R}^\infty$, but it is no longer part of the primary data.

§1.15 The $(\mathcal{V}_k, \oplus)$-cobordism group

The lecture specializes the lasagna story to

$$(\mathcal{V}_k, \oplus),$$

where $\mathcal{V}_k$ is the category or moduli object of finite-dimensional vector spaces under direct sum. One obtains groups denoted

$$\Omega_n^{\mathcal{V}_k, \oplus}.$$

The examples in low degree match algebraic K-theory:

$$\Omega_0^{\mathcal{V}_k, \oplus} \cong K_0(k) \cong \mathbb{Z},$$

$$\Omega_1^{\mathcal{V}_k, \oplus} \cong K_1(k) \cong k^\times.$$

Remark 1.15.1 — The slide summarizes the first two $(\mathcal{V}_k, \oplus)$-cobordism groups: degree 0 recovers dimension, and degree 1 recovers determinant. In formulas,

$$\Omega_0^{\mathcal{V}_k,\oplus} \cong K_0(k) \cong \mathbb{Z}, \qquad \Omega_1^{\mathcal{V}_k,\oplus} \cong K_1(k) \cong k^\times.$$

This is why vector-space-labelled circles are the K_1 analogue of the finite-set swap loop.

The degree-one case is especially helpful. A circle labelled by an automorphism $f \in \mathrm{GL}(V)$ represents a class, and the determinant gives its value in $k^\times$. A commutator becomes null-cobordant, matching the abelianization of $\mathrm{GL}_\infty(k)$.

This is the lasagna analogue of the finite-set story. In finite sets, a swap gives the nontrivial element of $\mathbb{Z}/2$. In vector spaces, an automorphism gives an element of $k^\times$.

§1.16 Generalized Pontryagin–Thom

The slide deck then states the central claim:

$$K_n(k) \xrightarrow{\cong} \Omega_n^{\mathcal{V}_k,\oplus} \qquad (n \geq 0).$$

This is presented as a generalized Pontryagin–Thom isomorphism.

Generalized Pontryagin–Thom Isomorphism

Recall that $K_0(k) = \mathbb{Z}$ and $K_1(k) = k^\times$. This is not just a coincidence:

Generalized Pontryagin–Thom Isomorphism (Claim)

For all $n \geq 0$, we have the isomorphism

$$K_n(k) \xrightarrow{\cong} \Omega_n^{\mathcal{V}_k,\oplus}$$

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Figure 1.5: From the lecture slides: the generalized Pontryagin–Thom isomorphism identifies $K_n(k)$ with a lasagna-cobordism group.

The analogy is:

classical/stable homotopy	algebraic K-theory side
$\pi_n^s(\mathbb{S})$	$K_n(k)$
Ω_n^{fr}	$\Omega_n^{\mathcal{V}_k,\oplus}$
framed manifolds	vector-space-labelled lasagnas

This is why the lecture moves from elementary arithmetic to Hopf fibration and then to intrinsic skein-lasagna. It is building a geometric model of algebraic K-theory.

§1.17 The K_2 slide and Steinberg relations

In degree two, the lecture recalls the classical presentation

$$K_2(k) \cong k^\times \otimes k^\times / \langle a \otimes (1-a) \rangle.$$

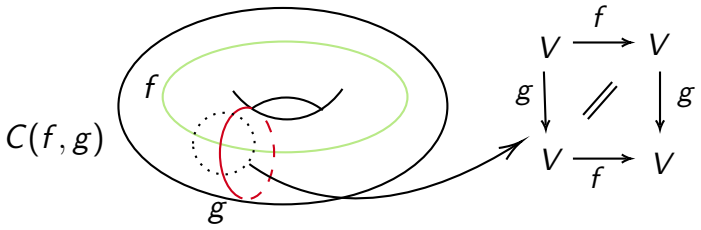
It then considers matrices $f, g \in \mathrm{GL}(k)$ with commutator

$$[f, g] = 1.$$

Such a commuting pair defines a two-dimensional lasagna $C(f, g)$.

$\Omega_2^{\mathcal{V}_k,\oplus}$ **and** $K_2(k)$

For matrices $f, g \in \mathrm{GL}(k)$ such that $[f, g] = 1$, one can define the following 2-lasagna $C(f, g)$:



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Figure 1.6: From the lecture slides: a two-lasagna $C(f, g)$ associated to a commuting pair.

The next slide gives a more concrete example using elementary matrices. For $i \neq j$, let

$$e_{ij}(a)$$

be the elementary matrix with a in the (i, j)-entry and ones on the diagonal. These matrices satisfy Steinberg relations, such as

$$[e_{ij}(a), e_{kl}(b)] = \begin{cases} e_{il}(ab), & j = k, \\ \mathbf{1}, & \text{otherwise.} \end{cases}$$

The slide then uses Hall–Witt-type commutator identities to produce a nontrivial proof that a certain commutator is the identity. This proof is not just algebraic decoration; it is a blueprint for constructing a bounding three-lasagna.

3-**Lasagna bounds** $C(e_{12}(a), e_{12}(b))$

For $i \neq j$, let the elementary matrices $e_{ij}(a)$ be

$$(e_{ij}(a))_{kl} = \begin{cases} 1, & \text{if } k = l, \\ a, & \text{if } k = i, l = j, \\ 0, & \text{otherwise.} \end{cases}$$

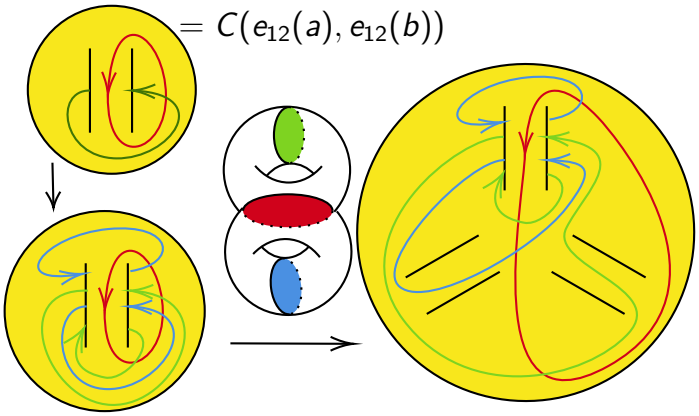
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Figure 1.7: From the lecture slides: elementary matrices, Steinberg relations, and the algebraic pattern behind the three-lasagna.

§1.18 The final lasagna diagrams

The final nontrivial slides display a Heegaard splitting and a boundary relation. They are not necessary for understanding the original meme, but they are essential for understanding where the lecture is going. The creation–swap–annihilation loop was the baby example. The lasagna diagrams are the research-level continuation of the same philosophy: algebraic K-theory classes are represented by geometric/cobordism-like objects.

Following this blueprint, we obtain the following Heegaard splitting to construct a lasagna:



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Figure 1.8: From the lecture slides: a Heegaard-splitting picture used to construct a three-lasagna.

At this point the lecture has travelled very far from the original equation. But the path is coherent:

equation as path
$\Downarrow$
finite signed configurations
$\Downarrow$
K-theory of finite sets
$\Downarrow$
stable homotopy and framed cobordism
$\Downarrow$
algebraic K-theory of fields
$\Downarrow$
intrinsic skein-lasagna as a geometric model.

§1.19 What the equation finally says

The equation

$$0 = 1 - 1 = -1 + 1 = 0$$

should be read as the name of a loop:

$$h : \varnothing \to \{+, -\} \to \{-, +\} \to \varnothing.$$

It is a loop at zero in the K-space of finite sets. It is not the identity loop. It represents the generator of

$$K_1(\mathrm{FinSet}) \cong \mathbb{Z}/2.$$

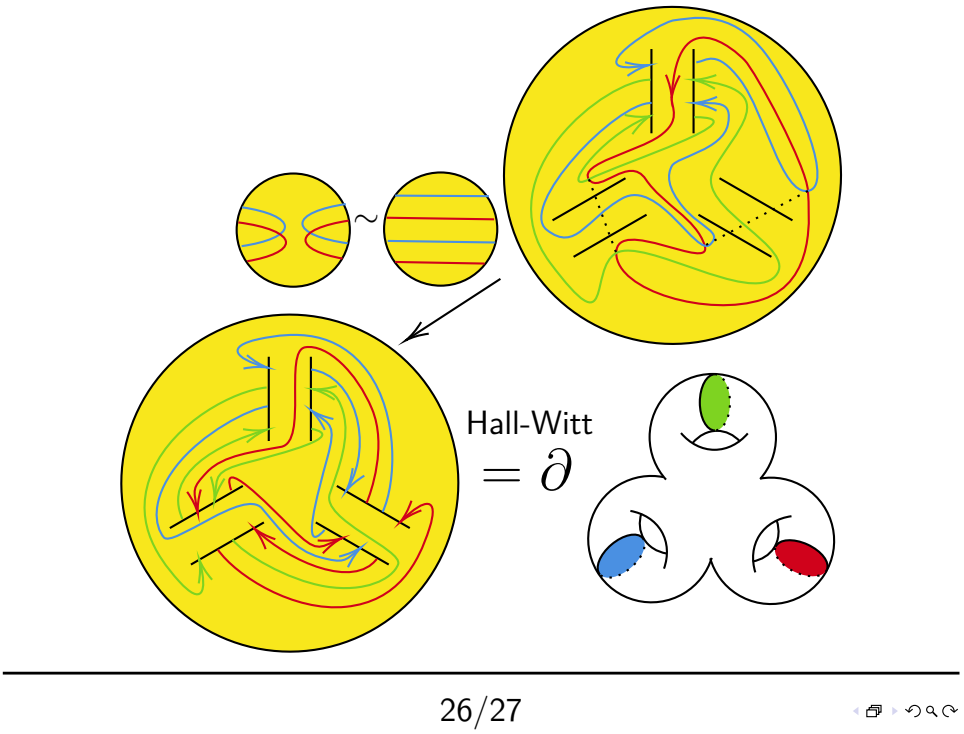


Figure 1.9: From the lecture slides: the boundary relation for the three-lasagna construction.

By Barratt–Priddy–Quillen, this group is

$$\pi_1^s(\mathbb{S}) \cong \mathbb{Z}/2,$$

and h corresponds to the stable Hopf class η.

This is why the Hopf fibration belongs in the same story as a signed finite-set calculation. The shared object is not an ordinary number. It is a stable homotopy class.